



Romina Saljooghian
Automation & AI Engineer

CONTACT

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TECHNICAL SKILLS

Programming
Python, MATLAB, SQL
ML & Deep Learning
PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, Hugging Face
LLM & Agentic AI
LangGraph, LangChain, OpenAI API, Anthropic API, Prompt Engineering
Backend & API
FastAPI, REST API, Pydantic, pytest
MLOps & Tools
MLflow, Hydra, Git, Linux, LaTeX
Control & Robotics
Simulink, MPC, LQR, Kalman Filter, PID, Kinematics, System ID

LANGUAGES

English — Fluent
Italian — A2
Persian — Native

CERTIFICATIONS

LangGraph Essentials
LangChain Academy, 2026
Automation Explorer
UiPath Academy, 2026
Python
freeCodeCamp, 2026
Relational Databases
freeCodeCamp, 2026

INTERESTS

Piano Puzzles

PROFILE

MSc graduate in Automation and Control Engineering with experience in deep learning, computer vision, agentic AI, and control systems. Built end-to-end ML pipelines from foundation-model orchestration to production REST APIs with LangGraph. Seeking roles in ML Engineering, CV, or Robotics AI.

EDUCATION

M.Sc. Automation & Control Engineering
Politecnico di Milano, Italy
Sep 2022 – Mar 2026
107 / 110
Thesis: Open-vocabulary scene graph generation for robotic manipulation using foundation models (CLIP, SAM, Grounding DINO, RAM) — MERLIN Lab.
B.Sc. Electrical Engineering (Minor: Control)
K.N.Toosi University of Technology, Tehran
Sep 2016 – Aug 2020
16.16 / 20
Selective Coursework: Neural Networks & Deep Learning · Control of Industrial Robots · Multiagent Systems · Software Engineering · Model Identification · Data-Driven Control · Autonomous Vehicles

PROJECTS

Scene Graph Generation for Robotic Manipulation
MSc Thesis · Python, PyTorch, CLIP, SAM, GPT API, Gemini API, Hydra
2025 – 2026
• Designed a modular open-vocabulary pipeline chaining 4 foundation models; **6× improvement** over end-to-end baseline (PredCls mR@100: 48.90%)
• Built VLM-based annotation pipeline (GPT + Gemini) with structured prompts, producing 246 spatial-relationship triplets for a novel robotics dataset
• Zero-shot cross-domain transfer to manipulation scenes (SGDet mR@20: 16.13%) with no fine-tuning
Compliance RAG Agent
Personal Project · Python, FastAPI, LangGraph, ChromaDB, BM25, OpenAI API
2026
• Built a RAG backend that indexes regulatory documents and answers questions with source citations via a **4-node LangGraph pipeline**
• Hybrid retrieval combining vector similarity (ChromaDB) with BM25 keyword search, merged using **Reciprocal Rank Fusion**; 21 automated tests
Visual QA Agent
Personal Project · Python, FastAPI, LangGraph, Anthropic API, OpenAI API, PyMuPDF
2026
• Built agentic backend answering visual questions about images and PDFs through a **4-node LangGraph pipeline** with conditional error routing at every stage
• Dual-provider VLM abstraction (Claude + GPT-4o); PDF-to-image conversion via PyMuPDF; 11 automated tests with full coverage
Plant Leaf Disease Classification
Course Project · Python, TensorFlow, FastAPI, MLflow, pytest
2024
• 5-model ensemble CNN achieving **86% accuracy**; async REST API with Grad-CAM explainability endpoint
• MLflow experiment tracking with nested runs; 6-module pytest suite covering API, data pipeline, and model registry
Time-Series Forecasting
Course Competition · Python, TensorFlow, Keras
2024
• Benchmarked 6+ architectures on 48K time series; Conv. LSTM with autoregressive decoding achieved **test MSE of 0.01**
CNC Machine — 2-DOF Pantograph Robot
Lab Project · MATLAB, Simulink, Quanser SRV02 Hardware
2024
• Deployed 9 controllers (PID, LQR, MPC, Kalman) on real hardware; MPC: **0.26 s settling, <0.5% overshoot**
• Near-perfect 2D trajectory tracking with KF + LQR; validated in both time and frequency domains on physical plant
Networked Control of a Vehicle Platoon
Course Project · MATLAB
2025
• LMI-based state-feedback controllers for a 4-vehicle platoon under 6 communication topologies; benchmarked centralized vs. distributed cost and convergence trade-offs

WORK EXPERIENCE

Software Developer
Torfeh Negar Holding, Tehran, Iran
Jan 2020 – Sep 2021
• Built production UI and backend modules for Holoo accounting software serving thousands of users
• Optimized SQL Server queries; resolved bottlenecks and stabilized nightly batch processes
Instrumentation & Control Intern
Tavan Gostare Zino, Tehran, Iran
Jun – Sep 2019
• Designed and tested industrial grounding and earthing systems for IEC standards compliance